Correcting Label-noise Corruption with lambda-Odds Ratio

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Abstract

Attribution in large-scale observational studies is often distorted by outcome misclassification and selection, which can inflate weak associations into significant risk factors or spurious "protective" effects. We present the lambda-Odds Ratio ($\boldsymbol{\lambda}$-OR), a misclassification-corrected odds-ratio estimator that (i) uses two ROC thresholds to select high-purity tails and (ii) corrects the observed $\mathbf{2} \times \mathbf{2}$ table via a minimal-feasible ridge inversion. Large-sample Wald intervals are obtained on the log scale, with total variance that includes the delta-method contribution from estimating selection-conditional error rates on a validation cohort. In simulation, the naive log-odds ratio exhibits attenuation and coverage collapse, whereas the corrected estimator remains effectively unbiased and sustains substantially higher coverage with only modest RMSE inflation. In EHR-scale applications to Alzheimer's disease and related dementias, idiopathic pulmonary fibrosis, and autism spectrum disorder, the method reduces inflated discoveries seen with naive odds ratios and SHAP attributions, while surfacing biologically credible modulators. These results provide a practical, defensible framework for attribution in biobank- and EHR-scale studies.

Keywords: spurious attribution; biobanks; electronic health records; label noise.

1 Introduction

Label noise hinders AI-driven EHR and biobank studies from recovering mechanistic insights: diagnostic and procedure codes that such studies mine, are generated for

billing workflow rather than research, and thus contain delayed, missed, and systematically biased assignments [1]. Several recently reported high-performing screening platforms achieve excellent discrimination, *e.g.* ZeBRA for ASD, IPF, perioperative cardiac events, and ADRD [2–5]. Other methods *e.g.*, RETAIN, provide attention-based saliency weights for interpretability in predictions [6]. But attention weights are not causal attributions, and do not correct misclassified outcomes in training. Consequently, attribution in large observational datasets faces two problems: systematic label noise and variance collapse, making chance associations appear significant or even spuriously protective [7].

In randomized trials, ORs have been historically used to reveal causal contrasts under minimal assumptions. However, in large scale observational settings, especially where we cannot identify cases precisely beyond looking for ICD code sets, classical ORs computed on raw exposure-outcome tables exaggerate significance. And this worsens as n grows. Post hoc attribution methods such as SHAP and LIME [8] are model-agnostic but do not address outcome misclassification [9], so instability under label noise and large-n inflation remains.

Classical label-noise corrections to OR mainly target small-sample misclassification [7] or require external gold-standard cohorts that are rarely available at scale.

In this work, we introduce the lambda-Odds Ratio (λ-OR), which corrects feature attribution by approximately recovering the true contigency table from the noisy exposure-outcome matrix using sensitivity-specificity value-pairs read from the ROC curve of a chosen pre-existing predictive model, thus explicitly accounting for misclassification before computing odds ratios. The *key insight* here is that if we have a good predictive model at hand (which is increasingly the case), the attribution problem can take advantage of the specifity-sensitivity trade-off achieved by the model to correct for expected label noise.

As a mathematical guarantee, λ-OR is shown to collapse under non-informative predictors (Theorem 2). Additionally, we can show that with baseline sensitivity p and specificity q, while a tiny row-wise error asymmetry $\delta > 0$ can make classical estimates of OR arbitrarily large when the $\mathrm{OR}_{\mathrm{true}} \approx 1$ (Lemma 4), λ-OR resists this potential inflation (Lemma 4 and Corollary 5). Likewise, mean-absolute SHAP can mis-rank a vanishing coefficient when feature variances differ substantially, a pathology formalized in the Gaussian case; λ-OR's collapse property prevents such mis-ranking when the underlying score is uninformative (Lemma 3 and Corollaries 5,7).

1.1 Motivation: Illustration of OR blow-up and SHAP mis-attribution

With $p = 0.80,\ q = 0.85,\ \delta = 0.10,\ \varepsilon = 0.01$, all error rates remain in $(0.70, 0.95)$, yet the OR diverges (Lemma 4 and Corollary 5):

$$\frac{\mathrm{OR}_{\mathrm{naive}}}{\mathrm{OR}_{\mathrm{true}}} = \frac{(0.90)(0.25)}{(0.70)(0.05)} = \frac{0.225}{0.035} \approx 6.43, \tag{1}$$

so $\mathrm{OR}_{\mathrm{naive}} \approx 6.5$. This demonstrates strong inflation from small differential errors under otherwise good (p, q).

For a SHAP example, with $|\beta_k| = 1$, $\sigma_k = 1$, $\varepsilon = 0.05$, and $\sigma_j = 30$, we obtain $S_j/S_k \approx 1.5$ (Corollaries 5,7), meaning a feature with a coefficient $20\times$ smaller appears 50% more important in SHAP attribution. Intuitively, SHAP scales importance by variance, so high-variance noise can dominate for weak effects.

To illustrate λ-OR applicability, Figure 2 compares three attribution strategies: odds ratios (left), mean SHAP values (middle), and λ-OR (right), applied to comorbidity data for IPF, ASD and ADRD, based on ZeBRA models. Each point corresponds to an ICD-coded condition, plotted by effect size on the x-axis and significance on the y-axis. While, applied to ADRD, IPF, and ASD cohorts, classical OR and SHAP attribute significance to an inflated fraction of diagnostic codes, with a striking number apparently "protective" (Fig. 2, columns 1,2), λ-OR suppresses these artifacts while preserving plausible modulators, yielding a far more interpretable attribution landscape (Fig. 2, column 3).

2 Methods

2.1 λ-OR Formulation

Consider retrospective dataset with binary case/control labels where we have on hand a predictive model f, and a corresponding ROC curve. We select two ROC thresholds: a high-specificity cut t_{hi} and a lower cut t_{lo}. Subjects with $f(X) \geq t_{\text{hi}}$ define exposed cases, and $f(X) < t_{\text{lo}}$ define controls, the middle region is excluded. As a rule of thumb, we use $t_{\text{hi}} \approx 97\%$ and $t_{\text{lo}} \approx 75\%$. The exposure $X \in \{0, 1\}$ is assumed error-free. The resulting 2×2 contingency table:

$$\tilde{T} = \begin{bmatrix} \tilde{a} & \tilde{b} \\ \tilde{c} & \tilde{d} \end{bmatrix}, \tag{2}$$

where $\tilde{a}$ is the count with $X = 1$ and labeled positive, and so on. Relative to the latent true label Y, define:

$$p_{\text{sel}} = \mathbb{P}(\tilde{Y} = 1 \mid Y = 1,\, I = 1), \qquad q_{\text{sel}} = \mathbb{P}(\tilde{Y} = 0 \mid Y = 0,\, I = 1),$$

estimated on a validation cohort, where $I = 1$ indicates inclusion by the threshold rules. The corresponding misclassification operator is

$$K = \begin{bmatrix} p_{\text{sel}} & 1 - p_{\text{sel}} \\ 1 - q_{\text{sel}} & q_{\text{sel}} \end{bmatrix}, \tag{3}$$

so that, in expectation,

$$\tilde{T} = TK^{\top}, \tag{4}$$

with T the unknown true count matrix.

2.2 Ill-posed Inversion and the Need for Regularization

Inversion of $K^{\top}$ to estimate the true contingency table via $T = \tilde{T}(K^{\top})^{-1}$ can result in infeasible solutions where the recovered cell counts are negative, which we address via a ridge-regularized inverse:

$$T^{(\lambda)} \triangleq \tilde{T}\,(K^{\top} + \lambda I)^{-1}, \tag{5}$$

where $\lambda > 0$ is chosen to ensure entries of $T^{(\lambda)}$ remain nonnegative and greater than a small threshold ϵ. The minimal such λ defines our feasible correction, and for the corrected contingency table

$$T^{(\lambda)} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \tag{6}$$

the corrected log-odds ratio (λ-OR) is defined:

$$\log \lambda\text{-OR}^{(\lambda)} = \log\left(\frac{ad}{bc}\right). \tag{7}$$

In practice, we use the smallest λ such that all recovered counts exceed a small threshold $\epsilon > 0$.

2.3 λ-OR Variance Estimation

Under the assumption of independent cell counts, the classical variance of the log-odds ratio is

$$\mathrm{Var}_{\mathrm{naive}}(\log \lambda\text{-OR}) = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}. \tag{8}$$

However, additional uncertainty arises from estimation of the selection-conditional misclassification rates p_{sel} and q_{sel}. Using the delta method, the extra variance is

$$\mathrm{Var}_{\mathrm{extra}} = \left(\left(\frac{1 - q_{\mathrm{sel}}}{(p_{\mathrm{sel}} + q_{\mathrm{sel}} - 1)^2} \right)^2 \frac{p_{\mathrm{sel}}(1 - p_{\mathrm{sel}})}{n_{\mathrm{val}}} + \left(\frac{1 - p_{\mathrm{sel}}}{(p_{\mathrm{sel}} + q_{\mathrm{sel}} - 1)^2} \right)^2 \frac{q_{\mathrm{sel}}(1 - q_{\mathrm{sel}})}{n_{\mathrm{val}}} \right), \tag{9}$$

where n_{val} is the size of the validation cohort used to estimate $(p_{\mathrm{sel}}, q_{\mathrm{sel}})$. The total variance is then

$$\mathrm{Var}_{\mathrm{total}}(\log \lambda\text{-OR}) = \mathrm{Var}_{\mathrm{naive}} + \mathrm{Var}_{\mathrm{extra}}. \tag{10}$$

The corrected Wald statistic is

$$Z = \frac{\log \lambda\text{-OR}^{(\lambda)}}{\sqrt{\mathrm{Var}_{\mathrm{total}}}},$$

with a two-sided p-value obtained from the standard normal tail,

$$p_{\mathrm{two}} = 2\Phi(-|Z|),$$

and $(1 - \alpha)$ confidence interval given by

$$\left[\log \lambda\text{-OR}^{(\lambda)} - z_{\alpha/2}\sqrt{\mathrm{Var}_{\mathrm{total}}},\ \log \lambda\text{-OR}^{(\lambda)} + z_{\alpha/2}\sqrt{\mathrm{Var}_{\mathrm{total}}}\right],$$

where $z_{\alpha/2}$ is the standard normal critical value.

2.4 Optimal selection of thresholds for estimating λ-OR

We define Λ-OR using a two-threshold design on the ROC curve. Specifically, we choose a high-specificity threshold t_{hi} to identify a positive cohort with very low false-positive rate, and a lower threshold t_{lo} to identify a negative cohort with very low false-negative rate. Subjects with scores $f(X) \geq t_{\mathrm{hi}}$ are treated as positive, those with

$f(X) < t_{\text{lo}}$ as negative, and the intermediate region is excluded. This construction yields selection-conditional sensitivity and specificity $(p_{\text{sel}}, q_{\text{sel}})$, which enter directly into the misclassification operator K.

Practical choices balance purity and sample size. In our applications, t_{hi} is set to achieve $\geq 97\%$ specificity, ensuring that the positive cohort is nearly free of controls, while t_{lo} is chosen so that specificity is $\approx$ 70–80%, ensuring a control cohort with negligible case contamination but adequate size. For rare outcomes, looser thresholds may be required to ensure sufficient sample.

Asymptotically, if the thresholds are taken arbitrarily close to the ROC axes (i.e., $t_{\text{hi}} \to 100\%$ specificity and $t_{\text{lo}} \to 100\%$ sensitivity), then $(p_{\text{sel}}, q_{\text{sel}}) \to (1, 1)$, the operator K approaches identity, and Λ-OR reduces to the classical odds ratio. Thus the two-threshold design interpolates between finite-sample robustness and the idealized noise-free limit.

To show this, let $S = f(\cdot)$ be our risk score (assumed to be calibrated), and define two threshold by t_{hi} (high specificity) and t_{lo} (low specificity). Let

$$H = \{S \geq t_{\text{hi}}\}, \qquad L = \{S < t_{\text{lo}}\}, \qquad I = \mathbf{1}\{H \cup L\}.$$

Write the selection–conditional sensitivity and specificity as

$$p_{\text{sel}} = \Pr(\tilde{Y} = 1 \mid Y = 1, I = 1) = \frac{\Pr(H \mid Y = 1)}{\Pr(H \cup L \mid Y = 1)}, \tag{11}$$

$$q_{\text{sel}} = \Pr(\tilde{Y} = 0 \mid Y = 0, I = 1) = \frac{\Pr(L \mid Y = 0)}{\Pr(H \cup L \mid Y = 0)}. \tag{12}$$

Assume class-conditional CDFs of S are continuous with strictly positive densities near the relevant tails. Further, let

$$\pi_H^{(y)} = \Pr(H \mid Y = y), \qquad \pi_L^{(y)} = \Pr(L \mid Y = y), \quad y \in \{0, 1\}.$$

Lemma 1 (Limiting selection)**.** *As the thresholds move to the ROC axes, i.e.,* $t_{\text{hi}} \to \sup S$ *and* $t_{\text{lo}} \to \inf S$*, we have*

$$p_{sel} \to 1, \qquad q_{sel} \to 1,$$

so the misclassification operator

$$K = \begin{bmatrix} p_{sel} & 1 - p_{sel} \\ 1 - q_{sel} & q_{sel} \end{bmatrix}$$

converges to the identity and Λ*-OR reduces to the classical odds ratio computed on the selected sample. However, the selected cell counts* (a, b, c, d) *satisfy*

$$\mathbb{E}[a] \asymp n\, \pi_H^{(1)} \Pr(X{=}1 \mid Y{=}1), \quad \mathbb{E}[b] \asymp n\, \pi_L^{(1)} \Pr(X{=}1 \mid Y{=}1),$$

with analogous expressions for c, d, so $\min\{a, b, c, d\} \xrightarrow{p} 0$ *as* $\pi_H^{(1)} \downarrow 0$ *and* $\pi_L^{(1)} \downarrow 0$. *Consequently,*

$$\mathrm{Var}_{naive}(\log \mathrm{OR}) = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \xrightarrow{p} \infty.$$

Proof By continuity of the class-conditional tails, as t_{hi} increases to the essential supremum, $\pi_H^{(0)} \to 0$ while $\pi_H^{(1)} > 0$ shrinks; similarly, as t_{lo} decreases, $\pi_L^{(1)} \to 0$ while $\pi_L^{(0)} > 0$ shrinks (follows from continuity). From the definitions, $p_{\mathrm{sel}} = \pi_H^{(1)}/(\pi_H^{(1)} + \pi_L^{(1)}) \to 1$ and $q_{\mathrm{sel}} = \pi_L^{(0)}/(\pi_H^{(0)} + \pi_L^{(0)}) \to 1$, hence $K \to I$ and the misclassification correction vanishes. Counts in each cell are binomial with expectations proportional to n times these tail probabilities; as the tails shrink, at least one term in $1/a + 1/b + 1/c + 1/d$ diverges, yielding the variance blow-up. □

2.5 Corollary (Bias–variance trade-off and interior optimum)

Let $J = p_{\mathrm{sel}} + q_{\mathrm{sel}} - 1$. For thresholds near the axes, the leading misclassification bias of $\log \mathrm{OR}$ is $O((1-J))$, while the variance scales as $O\big(1/n_{\mathrm{eff}}\big)$ with $n_{\mathrm{eff}} = n \cdot \min\{\pi_H^{(1)} + \pi_L^{(1)}, \pi_H^{(0)} + \pi_L^{(0)}\}$. Hence the mean-squared error admits the proxy

$$\mathrm{MSE}(\log \mathrm{OR}) \approx C_1 (1 - J)^2 + \frac{C_2}{n_{\mathrm{eff}}},$$

for constants $C_1, C_2 > 0$ depending on prevalences and exposure margins. As t_{hi} and t_{lo} approach the axes, $(1 - J) \downarrow 0$ (bias $\to 0$) but $n_{\mathrm{eff}} \downarrow 0$ (variance $\to \infty$), and conversely for looser thresholds. Therefore, there exists an interior choice of $(t_{\mathrm{hi}}, t_{\mathrm{lo}})$ that minimizes this proxy MSE, i.e., an *optimal* pair of cut-offs given the available sample size n.

2.6 Practical rule

Choose $(t_{\mathrm{hi}}, t_{\mathrm{lo}})$ to maximize J subject to a minimum n_{eff} constraint (e.g., each cell count exceeding a preset floor), or equivalently, minimize $(1 - J)^2 + \kappa/n_{\mathrm{eff}}$ for a user-specified κ reflecting desired precision.

3 Theoretical Guarantees

In the following we assume our risk scores are calibrated, i.e.,

$$\Pr(Y = 1 \mid S = s) = s, \ \forall s \in [0, 1] \tag{13}$$

Calibration ensures probabilistic interpretability but does not by itself determine the ROC geometry. Our primary separation measure is the two-sample Kolmogorov–Smirnov distance (Youden's J):

$$J = \sup_{t \in \mathbb{R}} \big|F_1(t) - F_0(t)\big|, \tag{14}$$

where F_1, F_0 are the class-conditional CDFs of S.

Lemma 1 (Extreme–strata prevalence gap) *Let $Y \in \{0,1\}$ have prevalence $\wp = \Pr(Y=1)$. Let S be a real-valued score. Fix $0 < \alpha < \frac{1}{2}$ and define*

$$R = 1: \; S \geq q_{1-\alpha}, \qquad R = 0: \; S \leq q_{\alpha},$$

where q_u is the u-quantile of the marginal *law of S. Then, writing $J = \sup_t |F_1(t) - F_0(t)|$,*

$$\Delta_Y \;:=\; \big|\Pr(Y=1 \mid R=1) - \Pr(Y=1 \mid R=0)\big| \;\leq\; \frac{2\wp}{\alpha}\, J. \qquad \text{(KS bound)}$$

Proof

$$\Pr(Y=1 \mid R=1) = \frac{\Pr(S \geq q_{1-\alpha}, Y=1)}{\Pr(R=1)} = \frac{\wp\,[1 - F_1(q_{1-\alpha})]}{\alpha}, \tag{15}$$

$$\Pr(Y=1 \mid R=0) = \frac{\Pr(S \leq q_{\alpha}, Y=1)}{\Pr(R=0)} = \frac{\wp\, F_1(q_{\alpha})}{\alpha}. \tag{16}$$

Hence

$$\Delta_Y \triangleq \big|\Pr(Y=1 \mid R=1) - \Pr(Y=1 \mid R=0)\big| = \frac{\wp}{\alpha}\,\big|1 - F_1(q_{1-\alpha}) - F_1(q_{\alpha})\big|. \tag{17}$$

Because the cut-points are defined on the marginal distribution of S, $F_0(q_{1-\alpha}) = 1-\alpha$ and $F_0(q_{\alpha}) = \alpha$. For any t we have $|F_1(t) - F_0(t)| \leq J$; evaluating at $t = q_{1-\alpha}$ and $t = q_{\alpha}$ and summing the resulting inequalities yields $|1 - F_1(q_{1-\alpha}) - F_1(q_{\alpha})| \leq 2J$. □

Theorem 2 (λ-OR collapse as $J \to 0$) *Let $X_j \in \{0,1\}$ be an exposure with $p_1 = \Pr(X_j{=}1 \mid Y{=}1)$, $p_0 = \Pr(X_j{=}1 \mid Y{=}0)$, and assume $\min\{p_0, p_1\} \geq \varepsilon > 0$. Let S be a calibrated score (Eq. (13)), define R by the α-quantile stratification above, and let $\wp = \Pr(Y=1)$. Suppose the observed outcome $\tilde{Y}$ has sensitivity p and specificity q, and that a ridge-regularized inverse-adjusted estimator $\log \lambda\text{-}OR_j^{(\lambda)}$ is formed as described above. Then*

$$\big|\log \lambda\text{-}OR_j^{(\lambda)}\big| \leq \frac{|p_1 - p_0|}{\varepsilon}\,\Delta_Y \leq \frac{2\wp\,|p_1 - p_0|}{\varepsilon\,\alpha}\, J. \tag{18}$$

In particular, if $J \to 0$ (the score becomes non-informative), then $\log \lambda\text{-}OR_j^{(\lambda)} \to 0$.

Proof Let $a = \Pr(X_j{=}1 \mid R{=}1)$, $b = \Pr(X_j{=}1 \mid R{=}0)$. From the PPV/NPV mixture identities, $a - b = (p_1 - p_0)\Delta_Y$. Since $b \geq \varepsilon$,

$$\Big|\log \frac{a}{b}\Big| \leq \frac{|a-b|}{\min(a,b)} \leq \frac{|p_1 - p_0|}{\varepsilon}\,\Delta_Y,$$

and Lemma 1 gives the J-bound. □

Corollary 3 (Lipschitz ROC implies λ-OR collapse as AUC $\to 0.5$) *Let $g(u) = \mathrm{TPR}(u) - u$ for $u = \mathrm{FPR} \in [0,1]$. If g is L-Lipschitz with $g(0) = g(1) = 0$, then*

$$\mathrm{AUC} \to 0.5 \Rightarrow \big|\log \lambda\text{-}OR_j^{(\lambda)}\big| \to 0$$

Proof

$$\mathrm{AUC} - 0.5 = \int_0^1 g(u)\,du \geq \frac{J^2}{L}, \Rightarrow J \leq \sqrt{L\,(\mathrm{AUC} - 0.5)}.$$

Consequently,

$$\left|\log \lambda\text{-OR}_j^{(\lambda)}\right| \le \frac{2\wp\,|p_1 - p_0|}{\varepsilon\,\alpha}\sqrt{L\,(\mathrm{AUC} - 0.5)} \xrightarrow[\mathrm{AUC}\downarrow 0.5]{} 0.$$

□

Because the ridge adds λ to every recovered count, our bound is loose, but the result stands.

Theorem 2 guarantees that λ-OR collapses under non-predictive stratification, suppressing significance inflation from weak models. We now contrast this behavior with classical odds ratio and mean-absolute SHAP, which can exhibit pathological amplification of vanishingly small true effects under common bias mechanisms.

3.1 Comparison against SHAP analysis

The SHAP approach to attribution may be briefly described as follows: For a prediction model $f : \mathbb{R}^d \to \mathbb{R}$ and an instance $x \in \mathbb{R}^d$, the *Shapley value* of feature j is

$$\phi_j(f, x) = \sum_{S \subseteq [d] \setminus \{j\}} \frac{|S|!\,(d - |S| - 1)!}{d!}\,\big[f\big(x_{S\cup\{j\}}\big) - f\big(x_S\big)\big], \tag{19}$$

the unique allocation of $f(x)$ that satisfies efficiency, symmetry, null-feature, and linearity axioms. A *global importance* for feature j is the *mean-absolute SHAP*

$$S_j := E_X\big[\,|\phi_j(f, X)|\,\big], \tag{20}$$

where the expectation is over the population distribution of X.

Linear Gaussian special case. If $f(x) = \beta^\top x$ and the components X_j are mutually independent Gaussian with mean μ_j and variance σ_j^2, then $\phi_j(f, x) = \beta_j(x_j - \mu_j)$ almost surely and

$$S_j = |\beta_j|\,\sigma_j\,\sqrt{2/\pi}. \tag{21}$$

Lemma 4 (Minimal inflation under differential error) *Let the true odds ratio be* $\mathrm{OR}_{\mathrm{true}} = 1 + \varepsilon$ *with* $|\varepsilon| \ll 1$. *Take baseline sensitivity* $p \in (0, 1)$ *and specificity* $q \in (0, 1)$. *Let the exposed row use* $p_1 = p + \delta,\ q_1 = q + \delta$ *and the unexposed row* $p_0 = p - \delta,\ q_0 = q - \delta$ *with* $0 < \delta < \min\{p, 1 - p, q, 1 - q\}$. *Then*

$$\frac{\mathrm{OR}_{naive}}{\mathrm{OR}_{true}} = \frac{(p + \delta)(1 - q + \delta)}{(p - \delta)(1 - q - \delta)}, \tag{22}$$

so $\mathrm{OR}_{naive} > \mathrm{OR}_{true}$ *for every* $\delta > 0$.

Proof The ratio follows by direct algebra from the transformed sensitivities and specificities. A first-order expansion in δ gives

$$\log \frac{\mathrm{OR}_{\mathrm{naive}}}{\mathrm{OR}_{\mathrm{true}}} \approx 2\delta\left(\frac{1}{p} + \frac{1}{1 - q}\right), \tag{23}$$

which is positive for all $p, q \in (0, 1)$. □

Corollary 5 (Odds-ratio blow-up with tiny row difference) *Fix $M > 1$. Choose $\varepsilon > 0$ arbitrarily small and set*

$$\delta \;\geq\; \frac{\log M - \log(1+\varepsilon)}{2\left(\frac{1}{p} + \frac{1}{1-q}\right)}. \tag{24}$$

With $p_1 = p + \delta,\ q_1 = q + \delta,\ p_0 = p - \delta,\ q_0 = q - \delta$ one has $\mathrm{OR}_{naive} \geq M$ even though $\mathrm{OR}_{true} = 1 + \varepsilon \to 1$.

Proof Immediate from Lemma 4. □

Corollary 6 (SHAP mis-ranking of vanishing coefficients) *Even if $|\beta_j| \ll |\beta_k|$, one may have $S_j > S_k$ whenever σ_j/σ_k is sufficiently large.*

Proof From (21), $S_j/S_k = (|\beta_j|\sigma_j)/(|\beta_k|\sigma_k)$. Choosing $|\beta_j| = \varepsilon$ and $\sigma_j = 30$ while $|\beta_k| = 1,\ \sigma_k = 1$ gives $S_j/S_k \approx 1.5 > 1$, establishing the mis-ranking. □

Corollary 7 (SHAP mis-ranking of a vanishing coefficient) *Let $\varepsilon > 0$ and $M = \max_{i \neq j,k} |\beta_i|$. If*

$$\sigma_j \;>\; \frac{\sigma_k\,|\beta_k|}{\varepsilon M + 1},$$

then $\mathbb{E}[\phi_j] > \mathbb{E}[\phi_k]$ and mean–SHAP preserves the true ordering of effects for the pair (j,k). Conversely, if

$$\sigma_j \;<\; \frac{\sigma_k\,|\beta_k|}{\varepsilon M + 1},$$

mean–SHAP can invert the ordering.

Proof Immediate from Lemma 4(b). □

4 Simulation Study

Design and data generation. We study estimation of a fixed $\theta = \log(\mathrm{OR})$ for a binary exposure $X \in \{0,1\}$ and outcome $Y \in \{0,1\}$ under outcome misclassification induced by a two-threshold selection on a calibrated predictive score for Y. We fix $\Pr(Y{=}1) = \pi$, a target log(OR), and the exposure prevalence $\Pr(X{=}1) = \varphi$. To generate (X,Y), draw $Y \sim \mathrm{Bernoulli}(\pi)$ and $X \mid Y{=}y \sim \mathrm{Bernoulli}(\theta_y)$ with (θ_0, θ_1) chosen to satisfy $\mathrm{OR} = \{\theta_1/(1-\theta_1)\}/\{\theta_0/(1-\theta_0)\}$ and $\varphi = \pi\theta_1 + (1-\pi)\theta_0$, yielding a latent 2×2 table $T = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

Label noise via two ROC gates. Independently of the data generation above, we construct two operating points on the ROC of the score $S = f(\cdot)$: a high-specificity threshold t_{hi} and a lower score cut t_{lo} chosen to yield high selection-conditional specificity q_{sel}. We include only the tails $H = \{S > t_{\mathrm{hi}}\}$ and $L = \{S < t_{\mathrm{lo}}\}$ (indicator $I = 1\{H \cup L\}$), which induce selection-conditional sensitivity and specificity

$p_{\text{sel}} = \Pr(\tilde{Y} = 1 \mid Y = 1, I = 1)$ and $q_{\text{sel}} = \Pr(\tilde{Y} = 0 \mid Y = 0, I = 1)$. This defines the misclassification operator $K = \begin{bmatrix} p_{\text{sel}} & 1 - p_{\text{sel}} \\ 1 - q_{\text{sel}} & q_{\text{sel}} \end{bmatrix}$ and, in expectation, $\tilde{T} = TK^{\top}$. Misclassification is non-differential with respect to X because I depends only on S.

Correction by stabilized inversion. Given $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d})$ and K, we recover a feasible table by ridge-regularized inversion

$$T^{(\lambda)} \;=\; \tilde{T}\,(K^{\top} + \lambda I)^{-1},$$

choosing the *smallest* $\lambda > 0$ such that all entries of $T^{(\lambda)}$ are nonnegative and exceed a small floor ε (feasibility rule). Writing $T^{(\lambda)} = \begin{bmatrix} \hat{a} & \hat{b} \\ \hat{c} & \hat{d} \end{bmatrix}$, the corrected estimator is

$$\log(\lambda\text{-OR}) = \log \hat{a} - \log \hat{b} - \log \hat{c} + \log \hat{d}.$$

Uncertainty uses the classical multinomial variance $1/\hat{a} + 1/\hat{b} + 1/\hat{c} + 1/\hat{d}$ plus the delta-method contribution from estimating $(p_{\text{sel}}, q_{\text{sel}})$ on a validation cohort, yielding $\text{SE}^2 = \text{Var}_{\text{base}} + \text{Var}_{\text{extra}}$ and Wald intervals for $\log(\lambda\text{-OR})$.

Metrics and grids. All performance is reported on the log scale against $\theta = \log(\text{OR})$. For replicates $r = 1, \ldots, R$, let $\widehat{\ell}_r^{(\text{naive})}$ be the naive log-OR from $\tilde{T}$ and $\widehat{\ell}_r^{(\lambda\text{-OR})}$ the corrected log-OR from $T^{(\lambda)}$. We report

$$\text{Bias} = \tfrac{1}{R}\sum_r (\widehat{\ell}_r - \theta), \text{RMSE} = \sqrt{\tfrac{1}{R}\sum_r (\widehat{\ell}_r - \theta)^2}, \text{Coverage} = \tfrac{1}{R}\sum_r \mathbf{1}\{\theta \in [\widehat{\ell}_r \pm 1.96\,\widehat{\sigma}_r]\}.$$

The primary grid increases misclassification by lowering p_{sel} while keeping q_{sel} high, with $m = 1 - p_{\text{sel}} \in [0.05, 0.20]$. The thresholds are chosen away from the ROC diagonal and the corners to balance purity and effective sample size. The same R is used in every cell, so coverage changes in steps of $1/R$.

Findings. As misclassification increases (See Fig. 1a-c), the naive log(OR) attenuates (bias drifts negative), RMSE inflates, and nominal 95% coverage collapses. In contrast, $\log(\lambda\text{-OR})$ remains essentially unbiased over the same range, with modest RMSE growth and substantially higher coverage. The eventual coverage decline at the largest m reflects intrinsic information loss as gates move toward the ROC diagonal together with the feasibility-enforcing ridge, rather than systematic miscalibration.

5 Application to Large EHR Databases

λ-OR addresses the principal failure modes of attribution at EHR/biobank scale: it collapses under non-predictive stratification (Theorem 2, Corollary 3), resists amplification of vanishing effects (Lemmas 4, and corollaries), and stabilizes correction under explicit label noise via ridge-regularized inversion across plausible (p, q) ranges. The practical upshot is that signals appearing ubiquitous and highly significant under classical OR or SHAP contract markedly when evaluated with λ-OR, while biologically

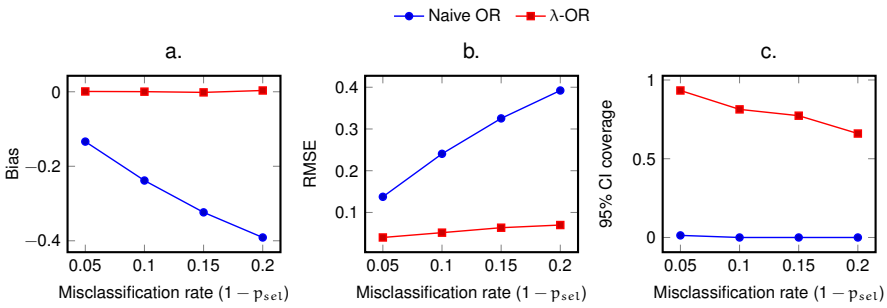

Fig. 1 Simulation results for bias, RMSE, and nominal 95% coverage of log OR under increasing misclassification rate $1 - p_{sel}$. Curves compare the naive estimator to the misclassification-corrected λ-OR using the minimal feasible ridge and ROC thresholds.

credible modulators persist (cf. Fig. 2). Intuitively, λ-OR computes exposure odds contrasts across calibrated risk strata and then corrects for outcome misclassification: if the score is uninformative the strata are indistinguishable and the contrast vanishes, whereas true modulators yield consistent separation that survives the correction.

Fig. 2 and Table 1 illustrates the implications of these guarantees: naïve odds ratios and SHAP values attribute significance to a wide cloud of diagnostic codes, with a substantial fraction appearing spuriously "protective.", while λ-OR retains only a narrow set of plausible modulators. Across ASD, IPF, and ADRD, the fraction of codes deemed significant by classical OR or SHAP is reduced several-fold, while the protective codes (often biologically implausible) are almost entirely eliminated.

For a specific example, λ-OR recovers established ADRD risk domains with strong variance-corrected signals: endocrine or metabolic (hyperparathyroidism, 14.118 [10]; hypo/hyperparathyroidism), sepsis (10.535 [11]), cardio- and cerebrovascular (hypertensive heart and kidney, 9.718 [12]; hypertension, 6.180 [13]; atrial fibrillation, 5.511 [14]; cerebrovascular, 5.395 [15]; atherosclerosis, 4.620 [16]), neurologic/psychiatric (traumatic brain injury, 8.841 [17]; major depressive disorder, 5.432 [18]; hearing loss, 4.693 [19]; sleep disorders, 2.277 [20]), and metabolic/respiratory/systemic conditions (type 2 diabetes, 4.434 [21]; COPD, 4.427 [22]; hyperlipidemia, 6.000 [23]; obesity, 2.791 [24]; age-related cataract, 3.930 [25]).

The small cluster of apparently "protective" codes lies in reproductive and gynecologic conditions (menstrual irregularities N91–N92, endometriosis N80.9, uterine/breast benign neoplasms D25–D26, cervical findings N84–N88, R87.61, R87.82), consistent with extensive evidence that cumulative estrogen exposure is neuroprotective. Women with longer reproductive spans (later menopause, later age at last childbirth) exhibit reduced dementia risk, whereas abrupt estrogen loss (surgical oophorectomy, premature menopause) is consistently associated with heightened risk. Mechanistically, estrogen supports synaptic and mitochondrial function, reduces β-amyloid aggregation, buffers oxidative stress, and dampens inflammation, all central to Alzheimer's pathogenesis [26, 27]. Thus, the appearance of these gynecologic codes as "protective" is biologically consistent with estrogen-mediated resilience against

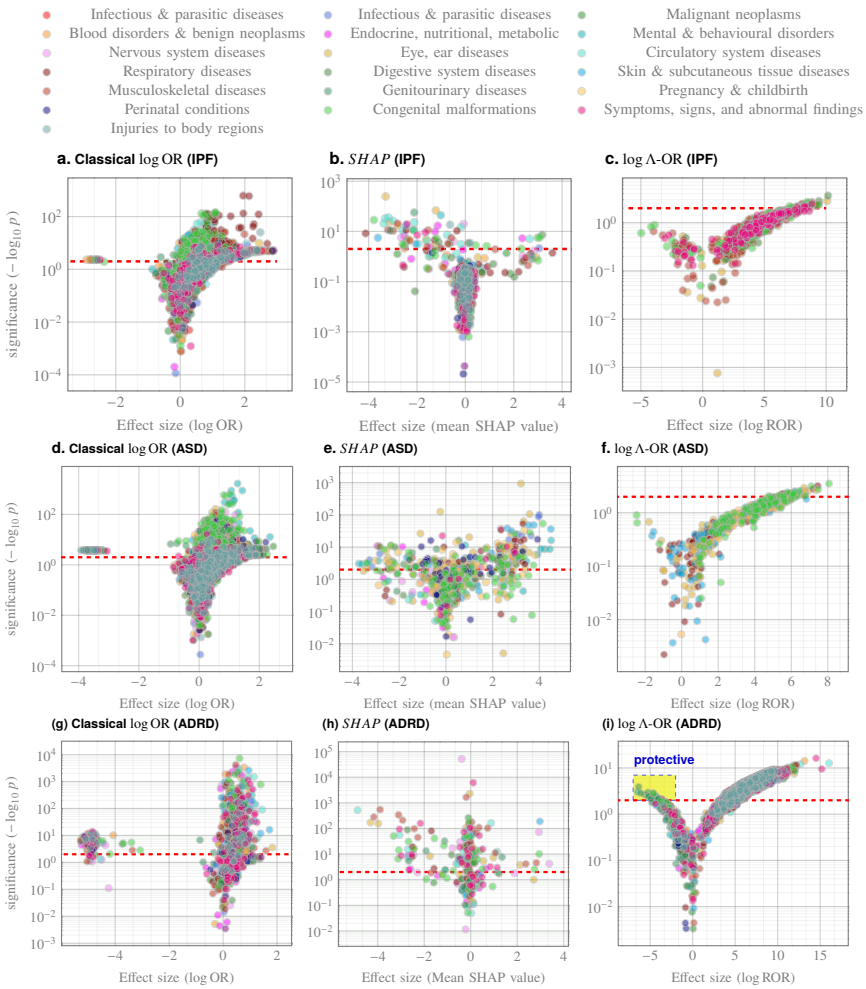

Fig. 2 Comparison of attribution methods (OR, mean SHAP, λ-OR) applied to comorbidity profiles for IPF, ASD, and ADRD. Each point is an ICD code (effect size on x; significance on y). OR and SHAP display pervasive significance and many spuriously "protective" codes due to label noise and large-n artifacts; λ-OR collapses this inflated cloud (Theorem S1), moderating significance and highlighting biologically credible modulators. Data: Merative Marketscan; λ-OR based on reported ZeBRA models [2, 3, 5]. Selection thresholds chosen at 97% specificity (cases) and 75% sensitivity.

neurodegeneration. A few seemingly unrelated codes also emerged, including attention-deficit hyperactivity disorder (F90), lateral epicondylitis (M77.11), and chronic viral hepatitis C (B18.2), wherein further investigation is warranted.

Concluding remarks

We have shown that a two–gate selection on a predictive score, combined with a minimal–feasible ridge inversion of the misclassification operator, yields a corrected log-odds ratio with simple Wald inference that removes classical attenuation from non-differential outcome noise. Across controlled simulations, the estimator remains essentially unbiased and sustains markedly higher coverage than the naive alternative over realistic misclassification levels, trading only modest variance. In EHR-scale applications (ADRD, IPF, ASD), it curbs inflated discoveries and spurious "protective" findings seen with naive ORs and model attributions, while surfacing biologically credible modulators.

6 Declarations

Ethics approval and consent to participate

Not applicable.

6.1 Consent for publication

All authors have reviewed and approved the final manuscript and consent to its publication.

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Availability of data and materials

The analyses rely on existing EHR- and biobank-derived resources accessed under data-use agreements that prohibit public release of individual-level records. De-identified contingency tables sufficient to reproduce all reported figures and tables, together with simulation seeds, will be made available upon publication.

A reference implementation and reproducibility scripts are available at https://doi.org/10.5281/zenodo.17196712 and https://github.com/zeroknowledgediscovery/lambda_or.

Competing interests

The authors declare no competing interests.

Authors' contributions

D.O. analyzed the data and drafted the manuscript. I.C. conceived the study, performed analyses, and co-wrote the manuscript. All authors reviewed and approved the final version.

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Algorithm 1: λ-OR Estimation with p-value

Input : Misclassified table $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d})$; ROC thresholds $(t_{\text{hi}}, t_{\text{lo}})$ with selection–conditional sensitivity p_{sel} and specificity q_{sel}; validation size n_{val}; ridge grid $(\lambda_{\text{start}}, \lambda_{\max}, \text{step})$; stability floor $\varepsilon > 0$.

Output: $(\log \lambda\text{-OR},\ -\log_{10} p)$ for the chosen $(t_{\text{hi}}, t_{\text{lo}})$.

Initialization:

$A \leftarrow \begin{bmatrix} p_{\text{sel}} & 1 - p_{\text{sel}} \\ 1 - q_{\text{sel}} & q_{\text{sel}} \end{bmatrix}, \quad \tilde{T} \leftarrow \begin{bmatrix} \tilde{a} & \tilde{b} \\ \tilde{c} & \tilde{d} \end{bmatrix}, \quad \lambda \leftarrow \lambda_{\text{start}}.$

Ridge inversion (minimal feasible λ):

while $\lambda \le \lambda_{\max}$ **do**

  $A_\lambda \leftarrow A + \lambda I.$

  counts $\leftarrow \tilde{T} \cdot A_\lambda^{-\top}$ // $A_\lambda^{-\top} = (A_\lambda^{-1})^\top$

  if $\min(\textit{counts}) \ge \varepsilon$ **then**

    break

  $\lambda \leftarrow \lambda + \text{step}$

if $\lambda > \lambda_{\max}$ **then**

  return (NaN, NaN)

Odds ratio:

Unpack counts into (a, b, c, d).

$\log \lambda\text{-OR} \leftarrow \log\left(\dfrac{ad}{bc}\right).$

Variance (base + delta for $(p_{\text{sel}}, q_{\text{sel}})$):

$\text{var}_{\text{base}} \leftarrow \dfrac{1}{a} + \dfrac{1}{b} + \dfrac{1}{c} + \dfrac{1}{d}.$

begin

  `// Delta-method partial derivatives (selection-conditional)`

  $\dfrac{\partial \log \lambda\text{-OR}}{\partial p_{\text{sel}}} \leftarrow \dfrac{(1 - q_{\text{sel}})(ad - bc)}{(p_{\text{sel}} + q_{\text{sel}} - 1)^2\, ad}, \quad \dfrac{\partial \log \lambda\text{-OR}}{\partial q_{\text{sel}}} \leftarrow \dfrac{(1 - p_{\text{sel}})(bc - ad)}{(p_{\text{sel}} + q_{\text{sel}} - 1)^2\, ad}.$

  $\text{var}_p \leftarrow \dfrac{p_{\text{sel}}(1 - p_{\text{sel}})}{n_{\text{val}}}, \quad \text{var}_q \leftarrow \dfrac{q_{\text{sel}}(1 - q_{\text{sel}})}{n_{\text{val}}}.$

  $\text{var}_{\text{extra}} \leftarrow \left(\dfrac{\partial \log \lambda\text{-OR}}{\partial p_{\text{sel}}}\right)^2 \text{var}_p + \left(\dfrac{\partial \log \lambda\text{-OR}}{\partial q_{\text{sel}}}\right)^2 \text{var}_q.$

$\text{SE} \leftarrow \sqrt{\text{var}_{\text{base}} + \text{var}_{\text{extra}}}.$

Significance:

$z \leftarrow \dfrac{\log \lambda\text{-OR}}{\text{SE}}.$

if $|z| \le 7$ **then**

  $p \leftarrow 2\big(1 - \Phi(|z|)\big)$ `// standard normal CDF` Φ

else

  $-\log_{10} p \leftarrow \dfrac{z^2}{2 \ln 10} - \log_{10}\big(\sqrt{2\pi}\,|z|\big).$

if $|z| \le 7$ **then**

  $-\log_{10} p \leftarrow -\log_{10}(p).$

Return:

return $(\log \lambda\text{-OR},\ -\log_{10} p).$

Table 1 Fraction of significant & protective codes for tested approaches

Type	DIS	ASD	IPF	ADRD
protective	λ-OR	0.000000	0.000000	0.001552
	shap	0.133609	0.019460	0.393471
	OR	0.021319	0.005457	0.023589
significant	λ-OR	0.071788	0.036725	0.863051
	shap	0.424242	0.024432	0.644419
	OR	0.365343	0.348754	0.624027